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Ensuring Wellbore and Equipment Integrity via Data-Driven Adaptive Limit Modeling for Pressure Management and Stuck Pipe Prevention

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Abstract

Drilling automation can greatly contribute to maintaining process and equipment safety during well construction, paving the way to operational and financial targets. Real-Time estimation and automated monitoring of operational limits for equipment and processes enables early detection of and reaction to potential hazards, e.g. stuck-pipe or over-pressure. This paper discusses a data-driven approach to determine limits for hook-load (HKLD) and stand-pipe-pressure (SPP), and how they are used to automatically prevent abnormal conditions.

As HKLD and SPP are constantly changing, the limits must adapt to them, which is a challenge. To achieve that, a statistical analysis of sensor data in real-time is performed to obtain safe operating envelopes (SOE) for both signals, where the outer boundaries of each SOE are henceforth referred to as limits. For HKLD these are derived by analyzing block movement, rig activity, and profile of the sensor signal over different time intervals. To include the drag in the system, pick-up (PU) and slack-off (SO) measurements are overlaid with the obtained limits which are updated after each transition to off-bottom. As soon as the HKLD exceeds a limit, a mitigation process is automatically triggered.

The calculation of SPP limits follows a similar approach. However, changes in flow rate are typically quite fast resulting in steep changes of SPP. Hence, determining adaptive pressure limits is even more challenging. Despite the rapid changes in the signals, the automation system can adjust the limits accordingly.

The algorithms have been verified and validated with simulated environments and in the field. Field tests have shown that both limits reliably adapt to the real-time signal. As a result, the monitoring algorithms detect overpulls as soon as the HKLD exceeds the upper limit and "set-downs" as soon as it falls below the lower limit.

The SPP limits are determined in scenarios with consistent flow rates and used to reliably monitor for potential over-pressures. The limits quickly adapt to steep changes in SPP which is critical when automating the circulation system.

In simulation environments where an automated drilling control system (ADCS) is configured by the drilling automation application, abnormal behavior of HKLD and SPP, such as overpulls or pack-offs, were

successfully mitigated automatically. This enhances the process of rig automation, increases safety and ensures the integrity of the equipment.

The presented approach yielded equivalent results to those of physics-based or machine learning-based approaches and can be made available in real time, which is crucial when an ADCS is used. In addition, other systems only account for static loads or pressure limits according to equipment specifications which does not consider operational constraints of the process. The presented approach provides dynamic limits derived directly from the monitored signals, thus reducing user interaction.

Introduction

Exploration and Production (E&P) operations are increasingly adopting low-cost digital technologies that offer dual benefits: reducing well construction capital expenditures by minimizing Non-Productive Time (NPT) and Invisible Lost Time (ILT), while also supporting Environmental, Social, and Governance (ESG) objectives, as illustrated in Fig. 1.

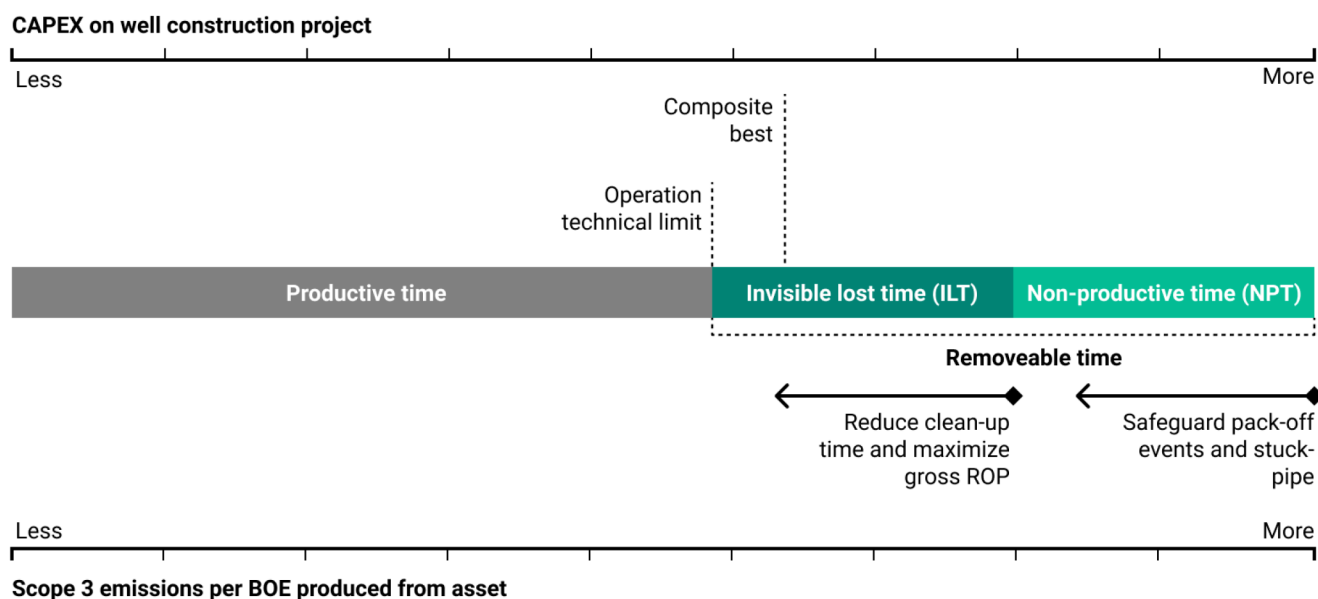


Figure 1—Productive and removeable time in well construction (Forshaw, et al. 2022)

One of the primary sources of NPT in well construction is stuck pipe—an incident that remains among the most challenging and costly problems in drilling operations. In such cases, the drillstring becomes immobile within the borehole due to mechanisms such as differential sticking, mechanical sticking, or packing off. These events not only contribute to NPT but may also result in equipment loss, requiring costly technical interventions that prioritize project continuity over Authorization for Expenditure (AFE). Moreover, stuck pipe incidents can severely impact safety and compromise wellbore integrity.

There are different approaches to identify, prevent and mitigate stuck-pipe events as summarized by (Montes, et al. 2025). In particular, one was described by (Katzmann, et al. 2024) where a data-driven model for real-time detection and prediction of stuck-pipe events is deployed at the edge using machine learning (ML). Such data-driven models can be enriched by coupling them with physics-based models that provide envelopes for wellbore and equipment integrity—such as digital twins that simulate wellbore and equipment behavior in real time (Arevalo, Hummes, et al. 2021), (Arevalo, May, et al. 2025), and drilling automation applications that monitor wellbore conditions (Ait Ali, et al. 2023), (Arevalo, Becker, et al. 2023). The hybrid modeling approach provides a more holistic safety environment where the wellbore, the equipment and the process are in scope of the protection algorithms.

This paper presents methods to estimate safe operating envelopes (SOE) for hookload (HKLD) and standpipe pressure (SPP) based on the operational context and the behavior of real-time signals—HKLD and SPP—to add a layer of protection oriented specifically to detect stuck-pipe precursors and compute limits for mitigating activities available in ADCS.

Method and Application

As HKLD and SPP are constantly changing, the limits must adapt to them, which is a challenge. To achieve that, a statistical analysis of sensor data in real-time is performed to obtain safe operating envelopes (SOE) for both signals, where the outer boundaries of each SOE are henceforth referred to as limits. For HKLD these are derived by analyzing block movement, rig activity, and profile of the sensor signal over different time intervals. As soon as the HKLD exceeds a limit a mitigation process is automatically triggered, as shown in Fig. 2.

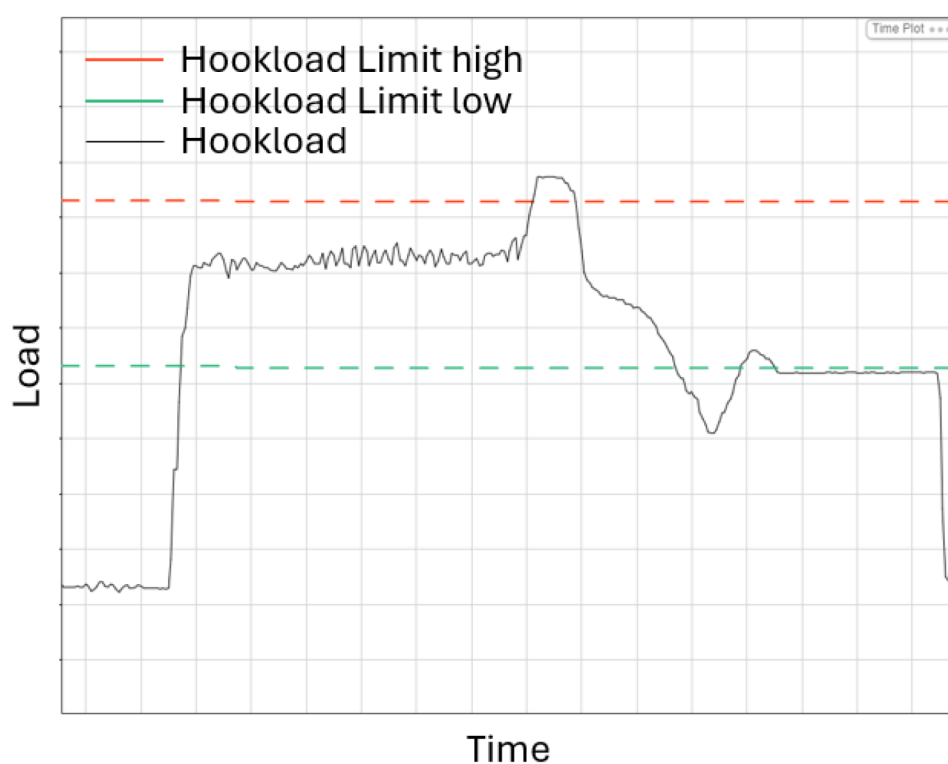


Figure 2—Example of Hookload exceeding Limits

The system applies HKLD limits if all the following conditions are satisfied:

- The value of HKLD remains above the specified threshold.
- The absolute bit velocity exceeds the specified threshold, which excludes non-informative static periods.
- The absolute bit acceleration stays below the specified threshold, as string acceleration can cause the expected high/low HKLD values to exceed the threshold.

Prior to a connection, the HKLD and SPP signals may fall below their respective lower thresholds, potentially triggering an alarm. To prevent false alarms before the connection, the system includes a dedicated delay mechanism and reconfirms the alarm based on the actual state.

Once the drillstring is off-bottom, data aggregation for limit calculations is enabled. First, a moving average of the hookload signal is computed. Then, the median of this moving average is determined. The limits are based on this median and are calculated as described in Eq. 1 and Eq. 2,

$$L_{HighStat} = H_{AvgMedian} + O_{UpperLim} \quad \text{Eq. 1}$$

$$L_{LowStat} = H_{AvgMedian} - O_{LowerLim} \quad \text{Eq. 2}$$

where:

- $L_{HighStat}$: Statistical hookload limit high
- $L_{LowStat}$: Statistical hookload limit low
- $H_{AvgMedian}$: Median of hookload average
- $O_{UpperLim}$: Offset to calculate upper limit
- $O_{LowerLim}$: Offset to calculate lower limit

Fig. 3 shows the real-time hookload signal and the resulting upper and lower limits by adding and subtracting the offsets. The figure also shows a spike in the hookload signal which is typical after a connection and during the transition out of slips. This spike is excluded from data acquisition and hence from the limit calculation by explicitly prolonging the transition out of slips in the algorithm.

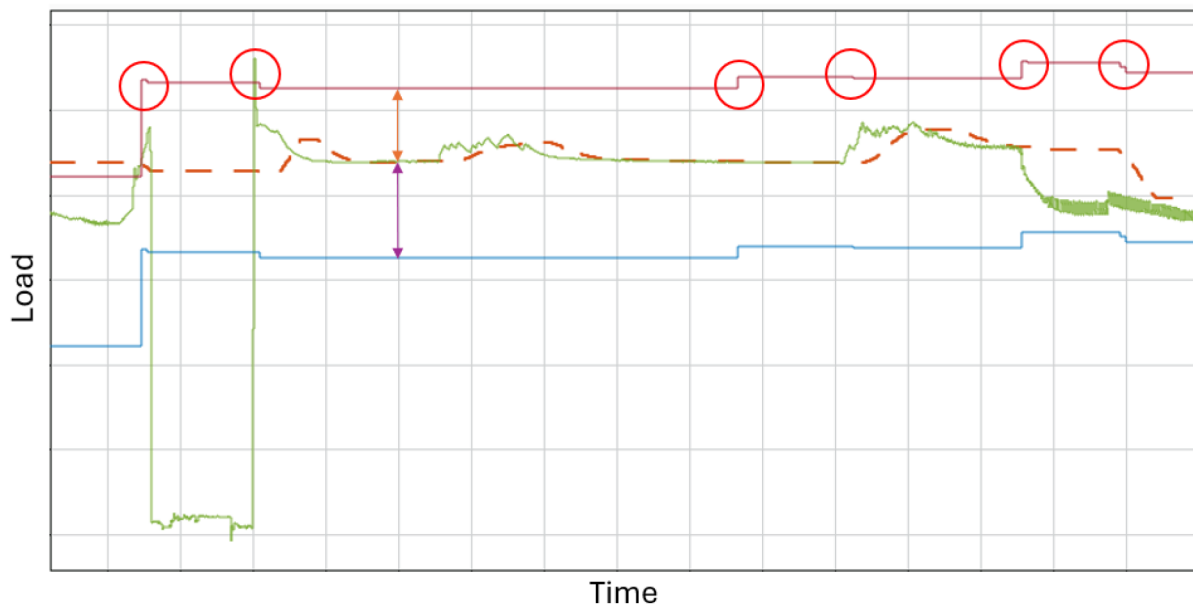


Figure 3—Hookload (Green), Median (dashed), Upper Limit (Red), Lower Limit (Blue), arrows indicating offset for limits, red circles indicating updates of limits

From an operational perspective, a non-fluctuating limit is required; therefore, additional conditions must be introduced before publishing the limits. To achieve this, block velocity and acceleration are monitored to first ensure stable block movement. If stability is confirmed, the limit is updated after a short delay. As shown in Fig. 3, fulfilled conditions are highlighted with a circle. Otherwise, the limit would follow the shape of the dashed median, which is not sufficiently steady when used in conjunction with an ADCS.

To account for drag in the system, pick-up (PU) and slack-off (SO) measurements are now overlaid with the calculated limits. For PU and SO sampling to occur automatically, specific conditions must be met. As a result, samples may only be aggregated every few stands. Since friction increases with depth, previously collected samples may become unsuitable. Therefore, PU and SO samples are only considered valid if they

were recently aggregated within a defined range of string movement. Assuming valid PU and SO samples are available, the hookload limits are determined as described in Eq. 3 and Eq. 4.

$$L_h = \max(L_{HighStat}, (S_{pickUp} + O_{Sample})) \quad \text{Eq. 3}$$

$$L_L = \min(L_{LowStat}, (S_{slackOff} - O_{Sample})) \quad \text{Eq. 4}$$

where:

L_H : Hookload limit high

L_L : Hookload limit low

S_{pickUp} : Calculated pick-up sample

$S_{slackOff}$: Calculated slack-off sample

O_{Sample} : Offset to calculate high and low limit

Fig. 4 shows where both limits are adjusted incorporating valid PU and SO samples.

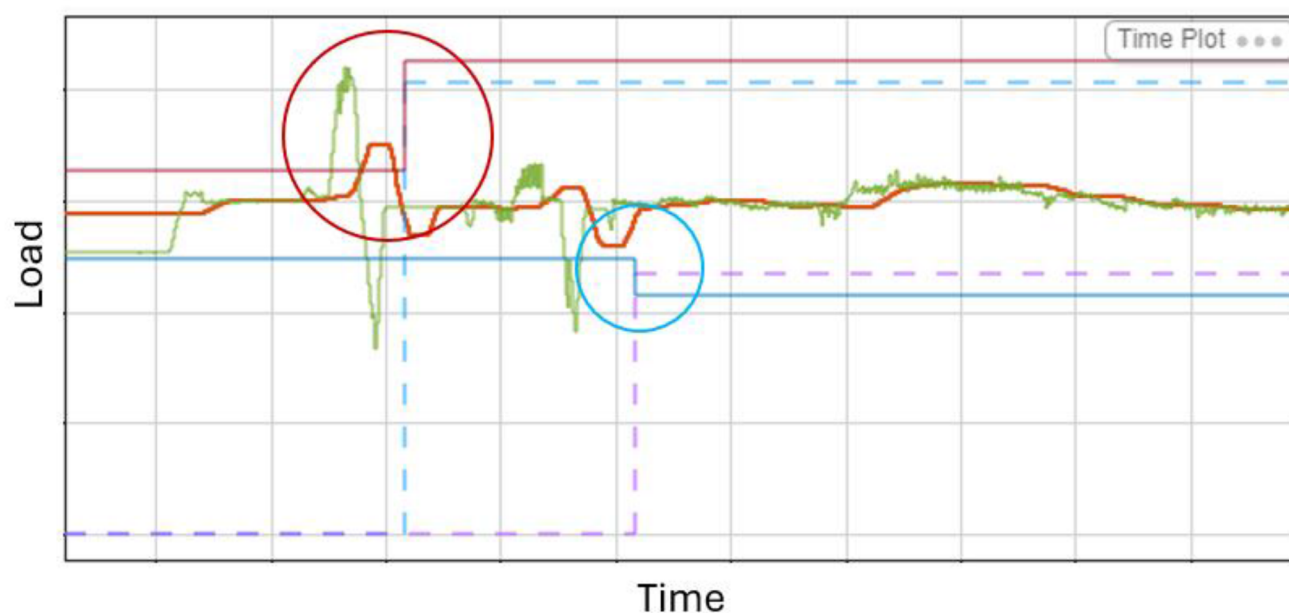


Figure 4—Hookload limits are adjusted with PU and SO samples (red circle indicates update of high with PU sample, blue circle indicates update of low limit with SO sample)

The approach for calculating SPP limits is similar to that used for hookload, but more complex. A calculated mean and an offset are used to define the limit. However, changes in flow rate are typically rapid, resulting in steep variations in SPP. This makes the determination of adaptive pressure limits particularly challenging. Despite these rapid signal fluctuations, the automation system is capable of adjusting the limits accordingly.

Additionally, downlink procedures must be excluded from consideration since the raw signals are disturbed. In this context, downlinking phases are flagged by the acquisition system, during which data aggregation is suppressed. For the purposes of the following description, it is assumed that no downlink is active.

To account for rapid changes in flow rate—and consequently in SPP—a simple statistical approach, such as the one used for hookload, is insufficient. As illustrated in Fig. 5, a change in flow rate is followed by a corresponding change in SPP. Using only a mean as the basis for limit calculation results in slow-reacting

limits. Reducing the window size used to calculate the mean, would make the limits more responsive, but also less smooth, which is undesirable.

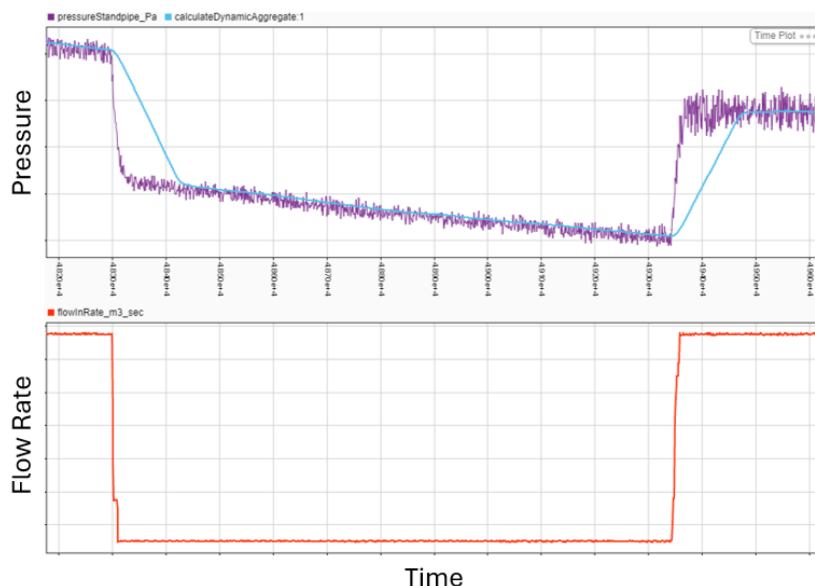


Figure 5—Top: SPP (Purple), SPP Mean (Blue), Bottom: Flow Rate (Orange). Bottom: Flow Rate is changing rapidly followed by changing SPP, Mean is only adjusting slowly

That is why a short-term and a long-term mean are used. To allow very fast adjustments of the limit the short-term mean is a weighted average. To automatically switch between short- and long-term, the algorithm constantly compares both, shown in Fig. 1. As soon as they are close enough to each other, the long-term is used. If they are deviating, the short-term is used.

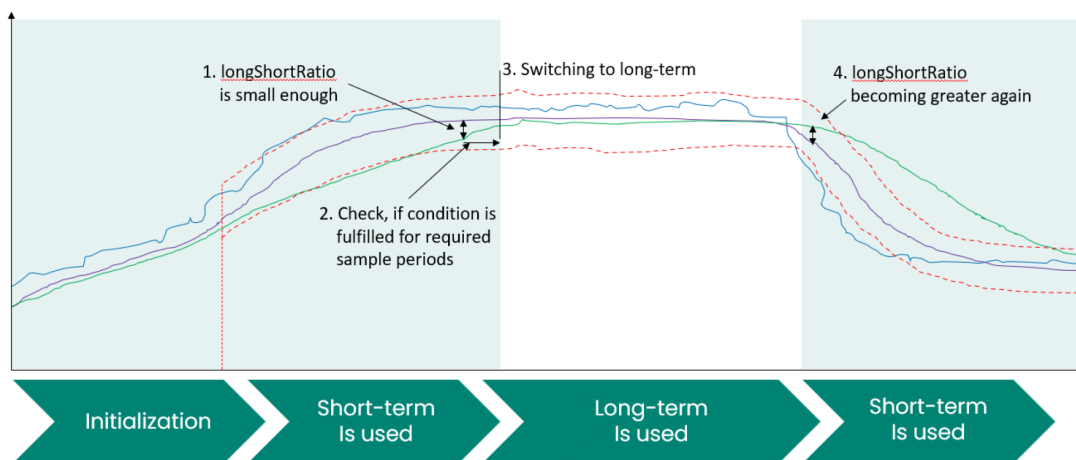


Figure 6—Switch between short- and long-term, Signal (Blue), Short-Term Mean (Purple), Long-Term Mean (Green), Signal Corridor for Limits (Red, dashed)

Using the mean as a basis the algorithm calculates an adjustable corridor around the signal. The corridor boundaries will serve as upper and lower limits. Regarding SPP, only the upper boundary is relevant in the current context. In a parallel process the median of the mean-based upper boundary is calculated to provide an even less fluctuating limit, if certain conditions are fulfilled.

Aggregating data to calculate the mean is only active when there is no ongoing pressure mitigation. Furthermore, it only receives data calculated when the flow rate and the pressure itself is in steady state, i.e. not changing. The median can be used, if one of the following scenarios are valid.

Scenario 1 – Flow rate is not steady, median within range:

- Difference between median and mean-based limit is smaller than required threshold
- Median limit is greater than lower boundary (calculated as described above)

Scenario 2 – Median is out of range but can still be used if:

- Difference between median and mean-based limit is smaller than required threshold
- Median limit is greater than lower boundary (calculated as described above)

The conditions to determine whether the median is within or without range are different, to incorporate hysteresis. As soon as the median becomes smaller than the previously calculated lower boundary and is deviating from the SPP mean, it is considered out of range.

Fig. 7 shows an example where the switches between median and mean based limits are performed. The red circle marks the point where the flow-in rate starts changing which results in an expected increase in pressure. This is as indicated by the increasing SPP mean and lower boundary in the first track. The reason why the SPP mean (green) is stepped, is because it is only published when the flow in rate has become steady. This happens during the ramp up for a short period and finally when it reaches its target value finally. That is why the mean is updated twice in the purple box. Since the window size for the median is greater than the one used for the mean, the median limit takes its time to be within range again. The green box indicates the point where the median limit reflects the two increases of the mean and where it is finally within range again, indicated by the flag on track three in Fig. 7.

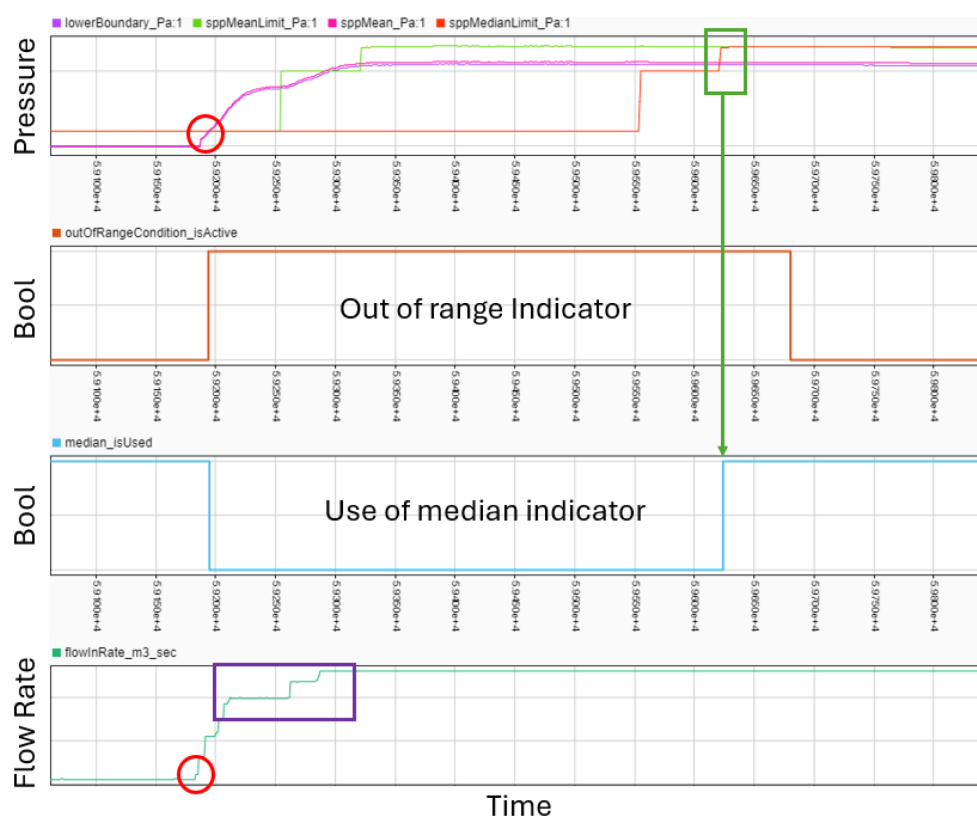


Figure 7—Example showing switching between mean and median limit: Flow rate starts increasing, SPP accordingly (red circle), flow rate becomes steady shortly and then reaches final value (purple box), median limit reaches condition where it can be used, although out of range condition is still active (green box)

The resulting limits as they can be published to the ADCS or be used in advisory mode are shown in Fig. 13. The red line represents the calculated limit. It is clearly visible that the limit is adapting quickly corresponding to the flow in rate increase but still not fluctuating.

Results

The presented signal processing approach yielded equivalent results to those of physics-based or machine learning-based approaches and can be made available in real time, which is crucial when an ADCS is used.

In addition, other systems only account for static loads or pressure limits according to equipment specifications which does not consider operational constraints of the process. Usually, these limits are too high to be used to detect, for example, pack-offs. Exceeding a mechanical design limit, or even a specified equipment operating limit, is usually beyond the limits which are meaningful for the ongoing process: the current process limits, that is, the limits placed on operational levels by the process itself. A comparison between the static rig pumps operating limits and the dynamics calculated process limit is shown in Fig. 8. The presented approach provides dynamic limits derived directly from the monitored signals, thus reducing user interaction.

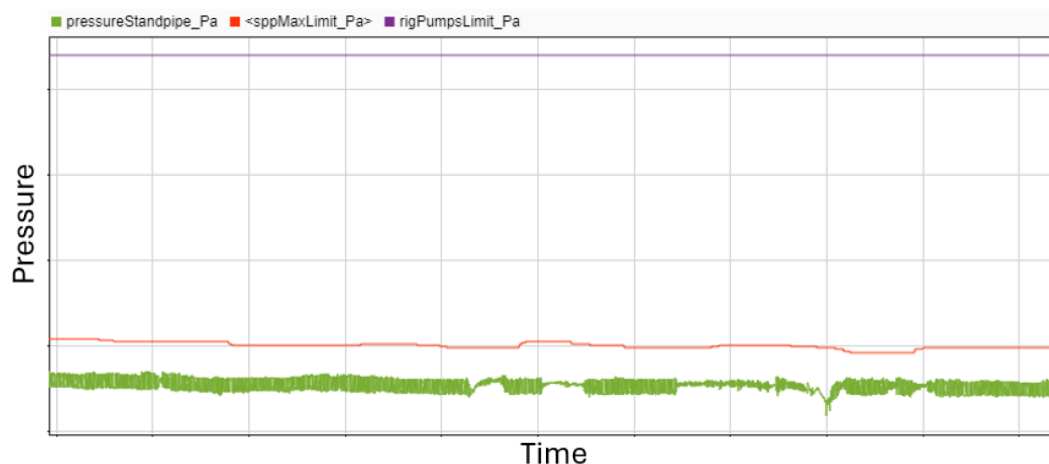


Figure 8—SPP (green) with data-driven Limit (red) compared to static Rig Pumps Limit (purple)

Fig. 9 shows a real example, where the hookload exceeded the upper limit. This triggered a mitigation procedure, in which the drillstring was lowered to reduce the load.

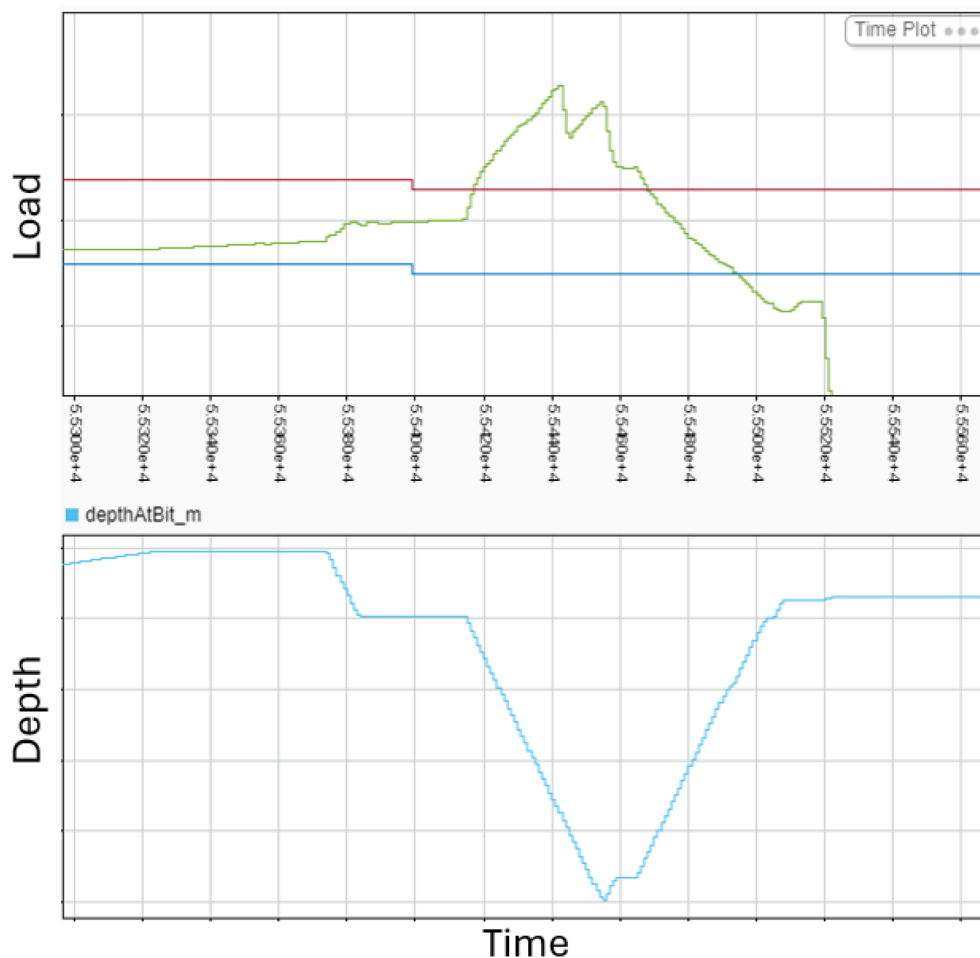


Figure 9—Example where high Hookload limit is exceeded and the drillstring is moved upwards, shown in second track, blue line

This was indicated in the rig control software as well, where the rig personal can clearly see, that the limit was exceeded, as illustrated in Fig. 10; and that a mitigation procedure was triggered, shown in the green box in Fig. 11.

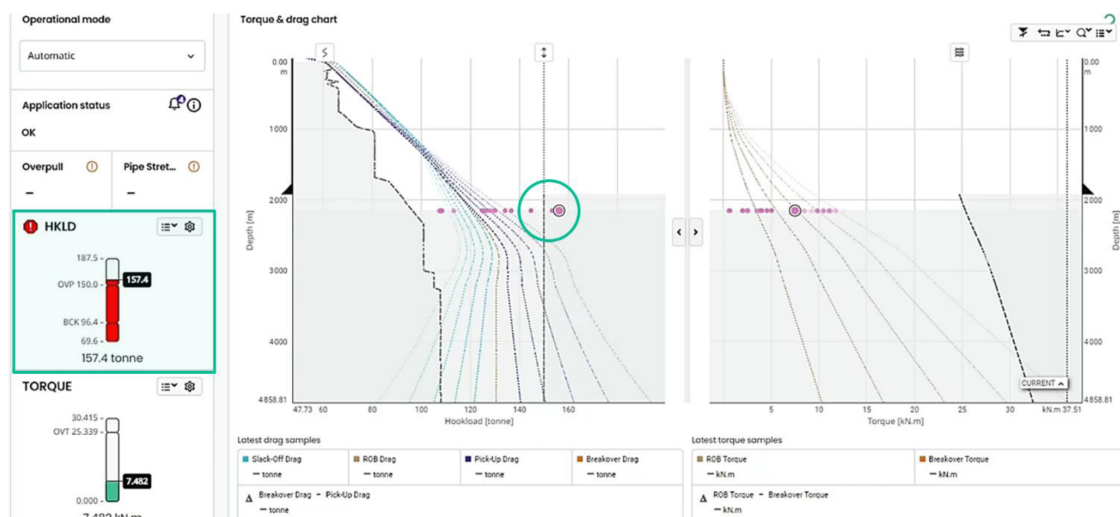


Figure 10—Dashboard with Hookload Gauge

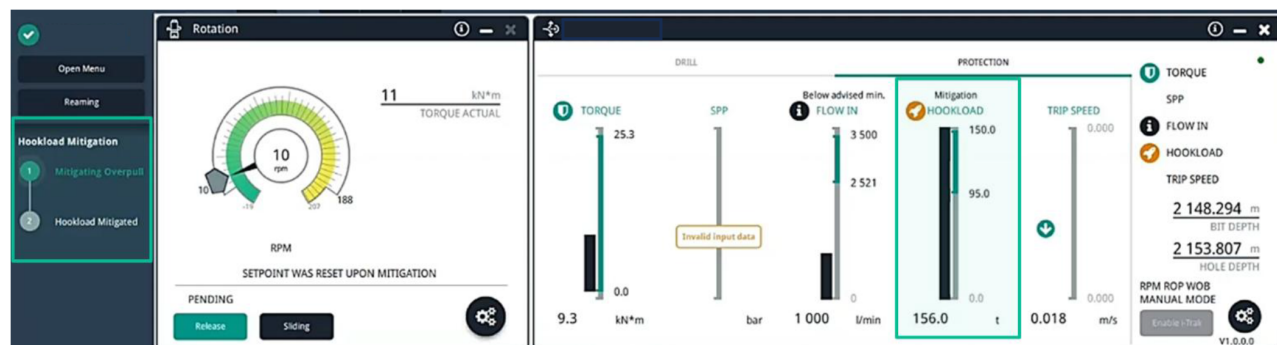


Figure 11—ADCS Application with Mitigation Indicator

The same scenario is shown in Fig. 12 when the lower limit is exceeded whereas the indicator in the ADCS application is the same.

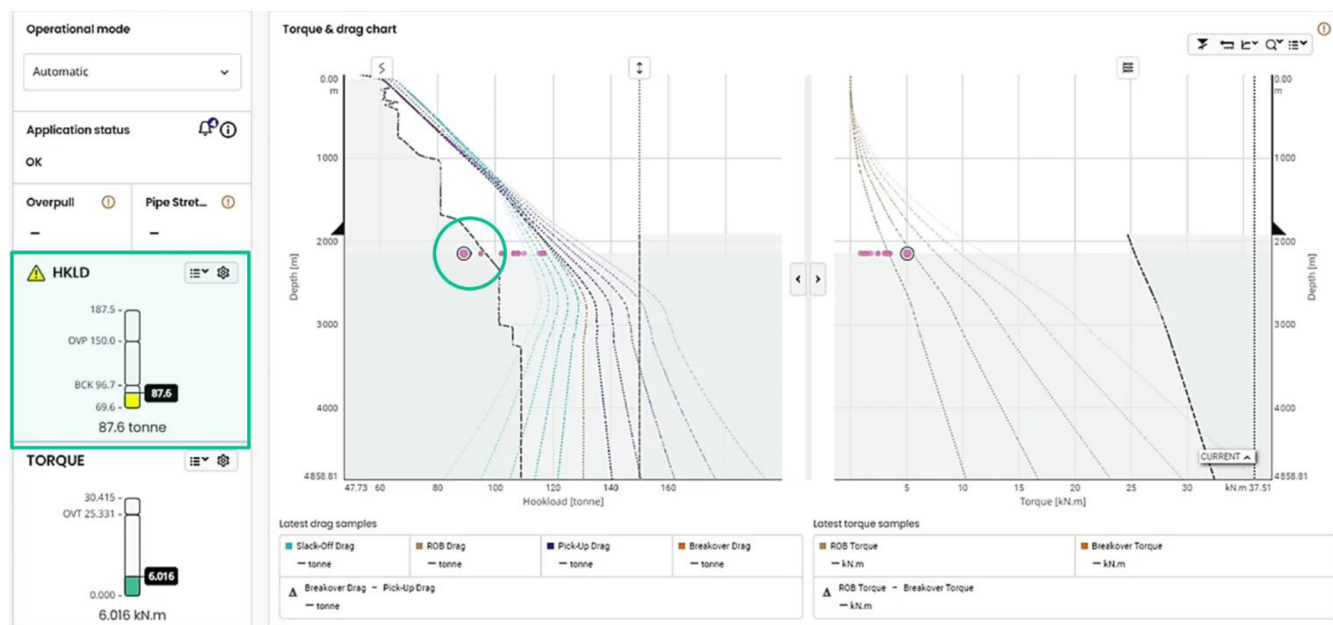


Figure 12—Dashboard with Hookload exceeding lower limit

The SPP limits are determined in scenarios with consistent flow rates and used to reliably monitor for potential over-pressures. The limits quickly adapt to steep changes in SPP which is critical when automating the circulation system. The resulting limits as they are published to the ADCS or be used in advisory mode are shown in Fig. 13. The red line represents the calculated limit. It is clearly visible that the limit is adapting quickly corresponding to the flow in rate increase but still not fluctuating. When the SPP rapidly increases, indicated by the red box in Fig. 13, the limit stays steady, which is the required behavior.

The basic condition for a pack-off is an increase of SPP while the flow in rate is steady. This is the case in the area highlighted by the red box. Fig. 14 shows the corresponding trigger indicating, that a pack-off is recognized by the algorithm.

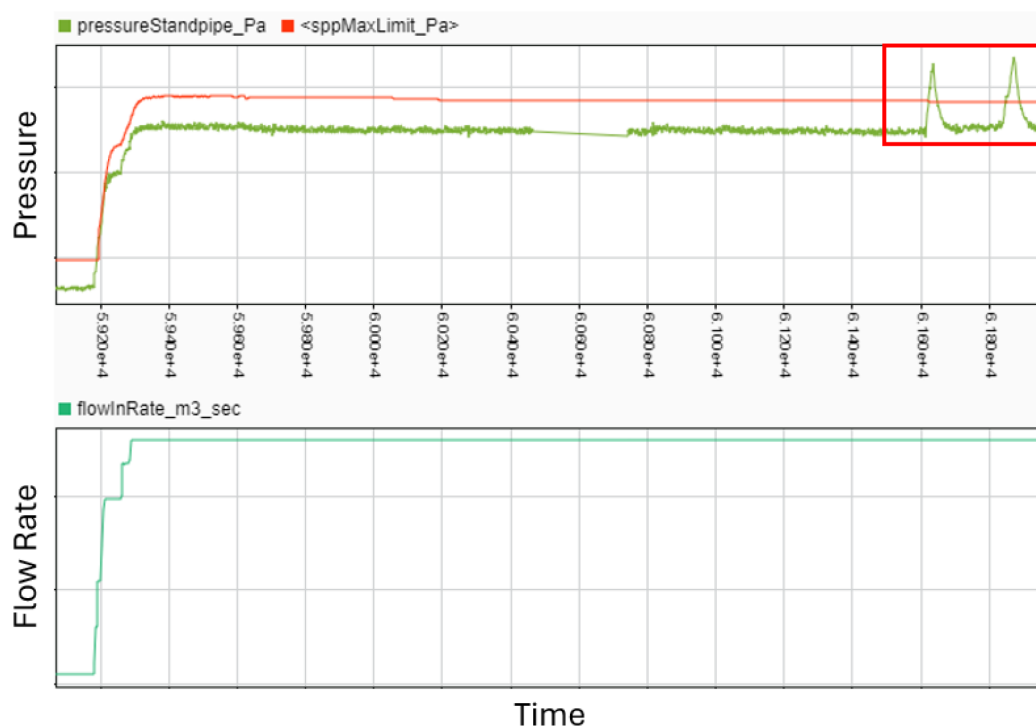


Figure 13—Pressure Limit for ADCS with pressure exceeding calculated limit (red box)

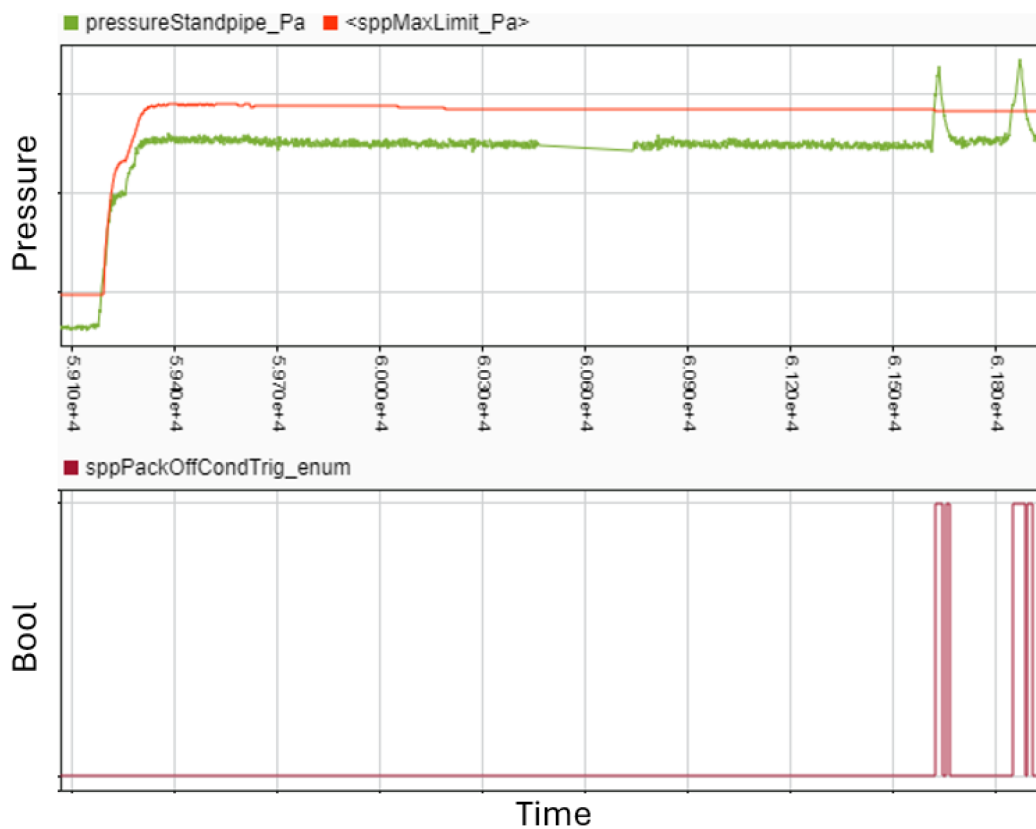


Figure 14—Pressure Limit is exceeded, System sets Trigger

In conjunction with an ADCS a pressure mitigation would be executed. Hence, the flow in rate would be reduced, until the pressure stabilizes again. Fig. 15 shows a borehole pressure screen from a drilling automation application running in a simulation environment. A quite steady limit for SPP is provided. When

this limit is exceeded, the ADCS applies a mitigation procedure to reduce the pressure. The mitigation procedure is highlighted in Fig. 15.

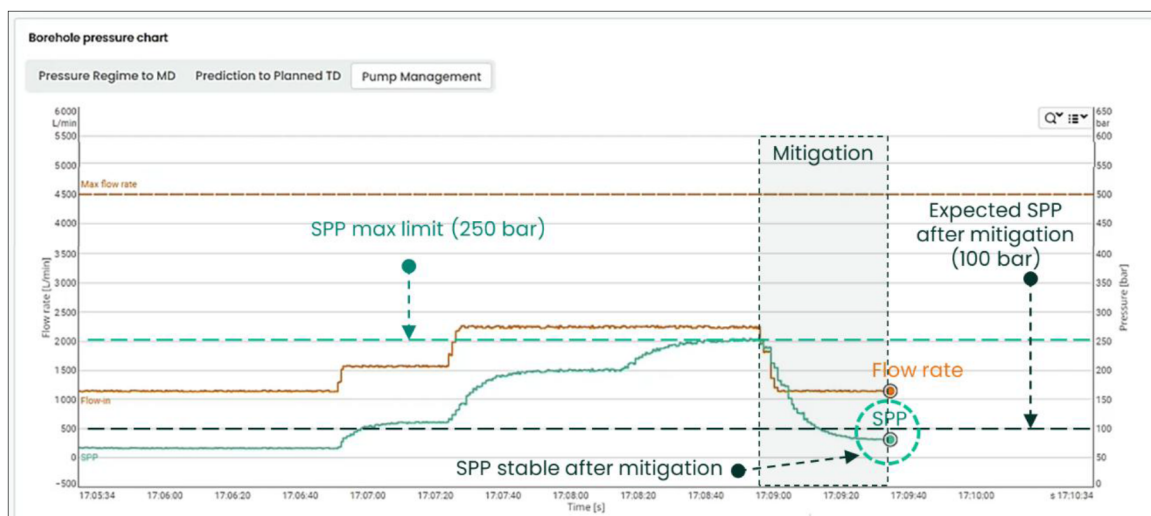


Figure 15—SPP limit used to mitigate Overpressure in Pack-off Scenario, calculated SPP Limit (Green, dashed), Flow Rate (Orange), Simulated Expected SPP after Mitigation (Black, dashed)

The automated pressure mitigation was not yet tested entirely in the field, since such scenarios are rare and cannot be deliberately provoked.

Conclusions

It has been demonstrated that adaptive, data-driven, limits function effectively. The hookload limits have successfully mitigated both overpull and set-down-weight scenarios in the field. However, when it comes to incorporating torque and drag samples, the validation process based on depth requires refinement. For the tests described and the results presented, samples were considered valid if they were aggregated within a specified depth range. Cases where the stand length exceeds the specified depth range—such as on very large rigs—can result in an undesired jump in the limit. Therefore, the validation process for torque and drag samples must be dynamic and adapt to the actual stand length.

Fig. 4 illustrates where pick-up and slack-off samples are incorporated. This leads to a jump in the limits, which, while not critical for the automation process, can undermine trust in the system due to the abrupt changes. A smoother transition would be beneficial in this context.

Additionally, the update frequency of the hookload limits needs to be increased. While updating them at the beginning of an off-bottom phase results in a steady limit for that section, scenarios involving a valid increase in hookload can trigger false positives. The logic should be refined to maintain steady limits while still adapting to valid increases or decreases in hookload.

The SPP limits already exhibit more dynamic behavior, as described in the results section. However, during ramp-up phases in flow rate, some false positives still occur. This is due to the time required for the buffers to update. As a solution, a long-term limit is planned, along with a detection mechanism that locks the long-term limit as soon as the SPP drops below a certain threshold—for example, when pumps are shut down. This would prevent the buffers from being influenced by very low SPP values.

Acknowledgements

The authors would like to thank Baker Hughes for permission to publish this work.

Nomenclature

ADCS	Automated drilling control system
AI	Artificial intelligence
BHA	Bottom hole assembly
FDIR	Fault detection, isolation and recovery
HKLD	Hookload
MD	Measured Depth
ML	Machine Learning
NPT	Non-productive time
OPC	Open platform communications
OPC UA	OPC unified architecture
RPM	Revolution per minute
RT	Real-time
SOE	Safe Operating Envelope
SPP	Standpipe pressure
UI	User interface

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